

VAPO: Value-Model-Based RL for Advanced Reasoning

ByteDance Seed · Base model: Qwen2.5-32B · No SFT data

Headline Result

60.4

AIME 2024 accuracy

>10 point lead over DAPO and DeepSeek-R1-Zero-Qwen-32B

Technical narrative: why value-model-based RL works for long chain-of-thought reasoning when bias, length heterogeneity, and sparsity are jointly addressed.

Why Value Models? Finer Credit Assignment for Long-CoT Reasoning

Motivation for value-model-based RL vs value-model-free alternatives

Value-model-free (GRPO/DAPO style)

- Trajectory-level advantage applied uniformly
- Weak token-level attribution in multi-step logic
- Harder to isolate where reasoning fails

Value-model-based (VAPO)

- Token-level credit assignment via value estimates
- More precise optimization for long reasoning paths
- Enables gains to 60.4 on AIME 2024

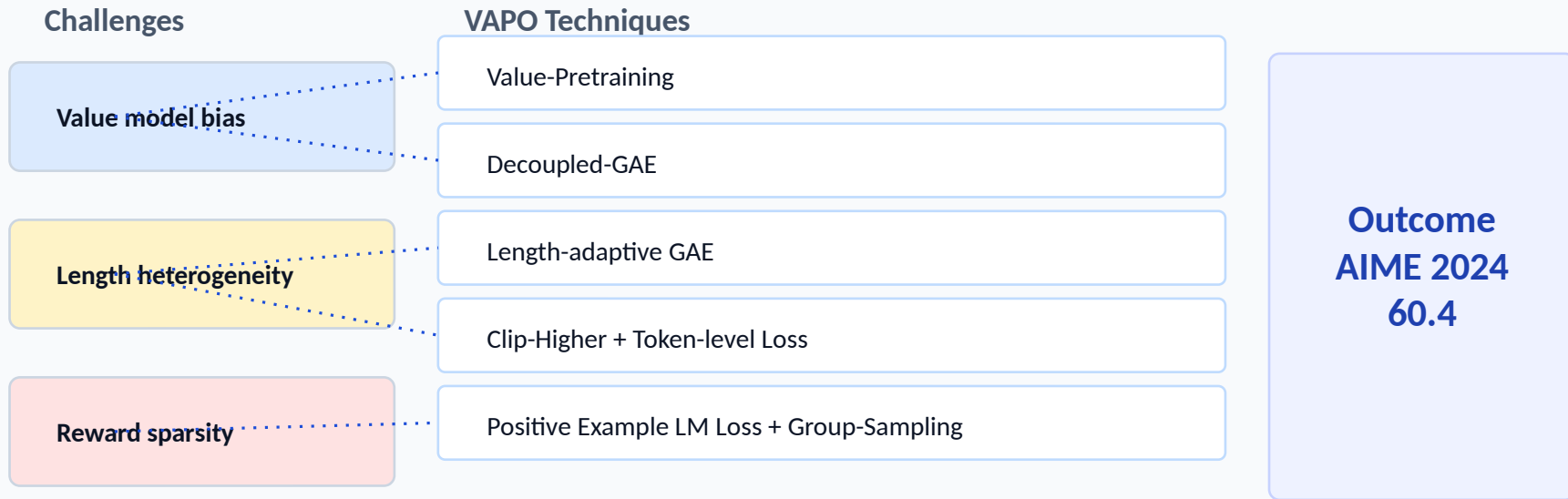
Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

Failure modes that previously limited value-model-based reasoning RL

1) Value Model Bias	2) Heterogeneous Lengths	3) Reward Sparsity
Long trajectories + bootstrapping amplify value estimation bias and destabilize policy learning.	Short and long responses need different bias-variance tradeoffs; fixed GAE λ is suboptimal.	Sparse outcome rewards in long CoT make effective credit propagation difficult.

VAPOR Framework: Integrated Solutions for Value-Model-Based Reasoning RL

Component diagram: each method mapped to addressed challenge(s)



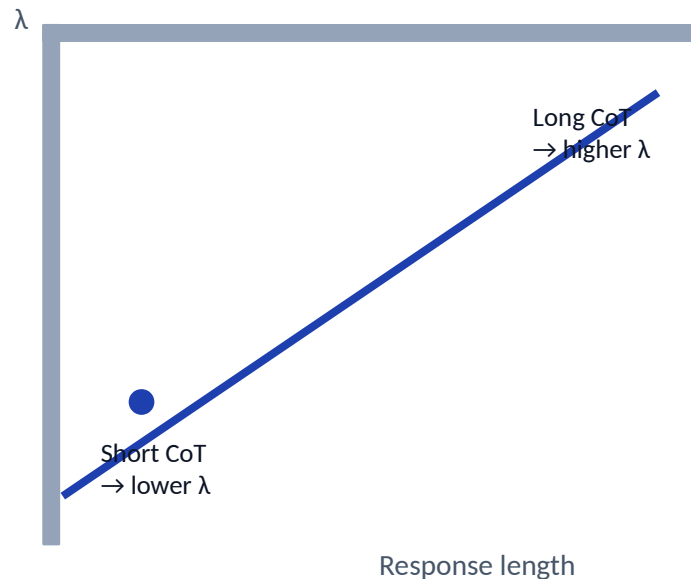
Length-Adaptive GAE: Dynamic λ for Variable Response Lengths

Short responses use lower λ (lower variance); long responses use higher λ (lower bias)

Conceptual mechanism

- Fixed λ fails under heterogeneous sequence lengths.
- VAPO adjusts λ by response length per sample.
- Improves bias-variance tradeoff across short/long CoT.
- Ablation: removing Length-adaptive GAE drops AIME from 60 to 45.

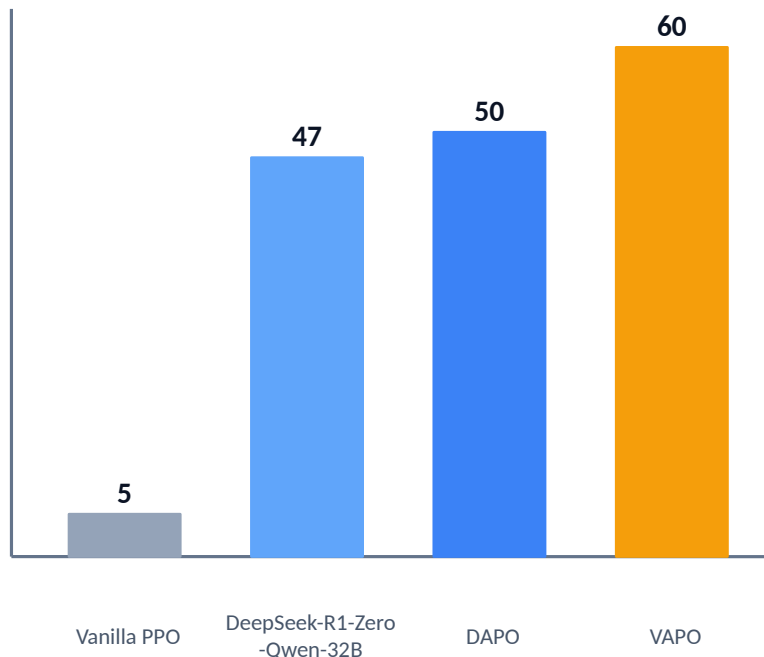
λ behavior by response length



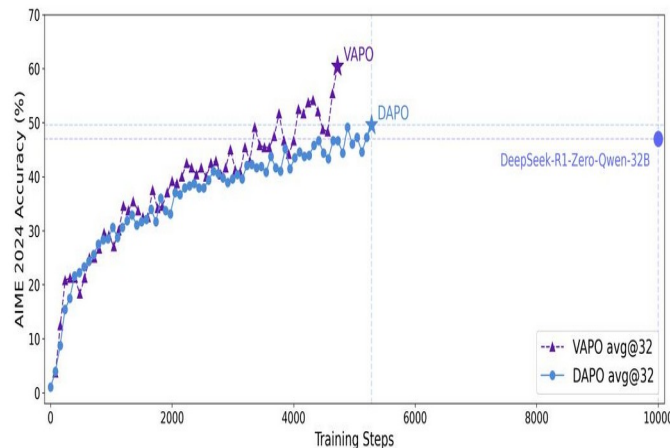
VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero

Main benchmark comparison and training curve reference (Figure 2)

AIME 2024 Accuracy (%)



Training curve (Figure 2)



VAPO reaches 50% at step 5,500; DAPO reaches 50% at step 10,000.

VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO

Figure 2: accuracy vs steps · Figure 3: mean response length

Figure 2: Accuracy curve

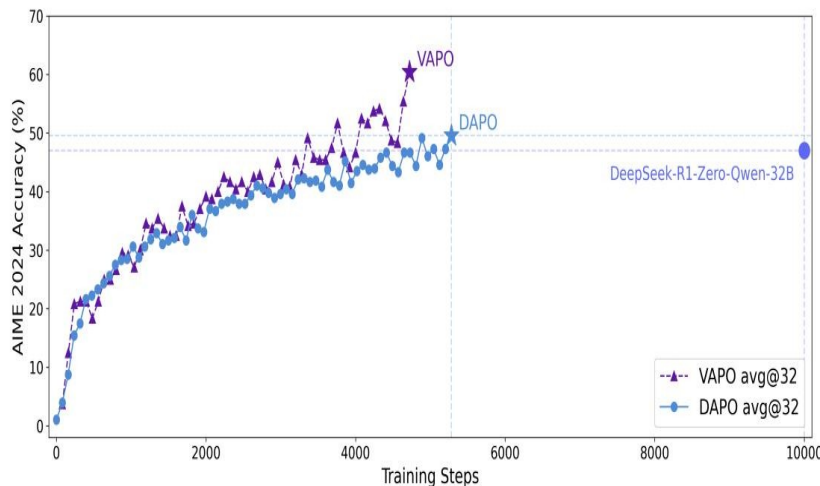
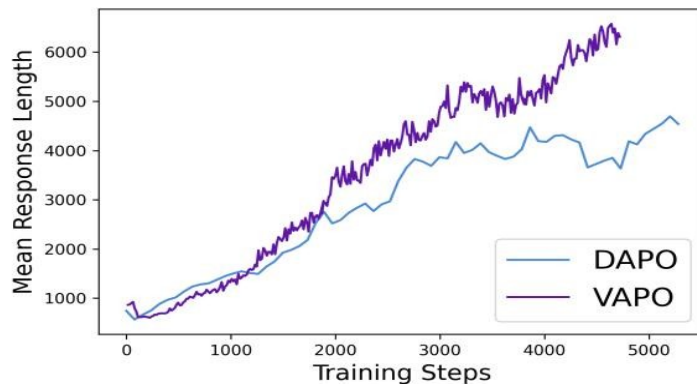


Figure 3: Response length



- VAPO reaches SOTA in ~5,000 steps.
- No training crashes across multiple runs.
- VAPO responses grow to ~6,000 tokens vs DAPO ~4,500.

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Annotated component contribution table (AIME 2024)

Configuration	AIME 2024	Δ vs Full
Full VAPO	60	Baseline
- Value-Pretraining	11	-49 (largest drop)
- Decoupled-GAE	33	-27
- Length-adaptive GAE	45	-15

Removing Value-Pretraining reduces performance from 60 to 11, indicating value-model-based RL is ineffective without strong value initialization.

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

- VAPO reaches 60.4 on AIME 2024, surpassing DAPO (50) and DeepSeek-R1-Zero-Qwen-32B (47).
- Value-model-based RL can outperform value-free methods when key failure modes are addressed systematically.
- Length-adaptive GAE is critical for heterogeneous response lengths (ablation: 60 $\rightarrow$ 45).
- Value-Pretraining is the highest-impact component (60 $\rightarrow$ 11 without it).
- VAPO is training-efficient and stable: SOTA in $\sim$ 5K steps, no training crashes across runs.

Implication: future advanced reasoning RL should treat value quality, adaptive credit assignment, and sparsity mitigation as a co-designed stack.